

The Climate Benefits of Improving Water Quality

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Abstract

Eutrophication enhances emissions of greenhouse gases (GHGs) from surface waters. Policies designed to ameliorate eutrophication by limiting nutrient loadings to surface waters can reduce these GHG emissions and, in turn, reduce future climate damages (e.g., from heat stress, sea-level rise, etc.)—yet this benefit has not been considered in benefit-cost analyses of water quality policies. We address this gap by using a set of linked watershed, lake, and aquatic GHG models to estimate emission reductions from a large-scale nutrient management program in America’s largest estuary, the Chesapeake Bay. The modeling system predicts reductions in chlorophyll-*a*, total phosphorus, and GHG emission rates in waterbodies throughout the watershed, but those in the southern portion of the watershed are predicted to exhibit greater reductions than those in the north, likely due to strong climate (e.g. ice-cover duration) and land-cover gradients across the domain. We estimate climate benefits from changes in GHG emissions from these water bodies of over \$300 million over the first 50 years of the program (2025-2075)—similar in magnitude to commonly quantified categories of water quality benefits. We then extrapolate our results to the third largest drainage basin in the world—the Mississippi-Atchafalaya River Basin—to estimate climate benefits of reduced GHG emissions from lakes and reservoirs in the basin resulting from a similarly stringent nutrient management policy. Our findings suggest that reductions in GHG emissions from nutrient management programs should not be overlooked when evaluating the societal benefits of such policies.

Core Ideas

- Improvements in water quality may lead to reduced greenhouse gas (GHG) emissions from surface waters.
- We estimate the Chesapeake Bay TMDL will reduce CO₂, CH₄, and N₂O emissions from lakes in the watershed.
- The monetized value of reduced GHG emissions are similar to other commonly estimated benefit categories.
- Benefit-cost analysis of water quality regulations should account for the value of avoided climate damages.

1 INTRODUCTION

Nutrient over-enrichment of inland and coastal waters is one of the greatest threats to water quality worldwide (Damania 2019; United Nations Environment Programme 2021). In the United States, over half of the nation's stream miles, and an estimated 40% of its freshwater lakes and reservoirs, are impaired because of excess nitrogen and phosphorus (Stoddard et al. 2016; U.S. EPA 2020, 2022). More than three-quarters of assessed coastal waters exhibit symptoms of eutrophication such as excess algal growth, low dissolved oxygen, and loss of submerged aquatic vegetation (Selman et al. 2008). Watershed management plans can significantly reduce nutrient pollution from agriculture, industry, urban stormwater, and wastewater treatment plants, but can be costly to implement. Benefit-cost analyses (BCA) of such management plans often find that costs exceed the monetized benefits—possibly because important benefit categories are overlooked or hard to measure (Keiser et al. 2019). These BCAs tend to monetize the economic benefits from improved recreational experiences and reduced drinking water treatment costs (Griffiths et al. 2012) but overlook other benefits such as improvements to human health via cleaner drinking water and enhancements to groundwater quality, coastal ecosystems, and the atmosphere (Jones 2019; Keiser et al. 2019).

One benefit of reduced nutrient pollution that has not been accounted for in BCAs of watershed management policies is the climate benefit resulting from reductions in greenhouse gas (GHG) emissions from surface waters. Nutrient pollution can affect GHG emissions from surface waters by controlling the

availability of carbon or oxygen to the microbial community. Algal blooms, which can be stimulated by nutrient pollution, generate carbon that is readily metabolized by microorganisms and has been shown to stimulate methane (CH_4) production in aquatic ecosystems (West et al. 2012; West et al. 2015). Algal-derived carbon can also promote oxygen depletion in aquatic environments by stimulating aerobic microbial respiration. This is important because several microbial processes that produce GHGs (e.g., methanogenesis, anaerobic denitrification) can only occur in low oxygen environments. Furthermore, nitrous oxide (N_2O) emission rates have been shown to correlate with algal blooms (Mengis et al. 1997; Wang et al. 2006) likely because 1) some types of algae can directly produce N_2O (Weathers 1984; Burlacot et al. 2020), and 2) carbon and ammonium released from senescent algal cells can stimulate denitrification and nitrification, types of microbial nitrogen transformation processes that can produce N_2O (Chen et al. 2016).

Models that predict GHG emissions from waterbodies based on levels of nutrient pollution, algae abundance, and other factors have been used to estimate global GHG emissions from lakes and reservoirs (DelSontro et al. 2018), examine the effect of reservoir water-level management on GHG emissions (Waldo et al. 2021b), and develop projections of global CH_4 emissions from lakes and reservoirs over the next century (Beaulieu et al. 2019). However, only one study to date has used surface water GHG models to estimate climate benefits of reducing nutrient loading to freshwater systems (Downing et al. 2021). That study estimates reductions in CH_4 emissions from Lake Erie, one the U.S. Great Lakes, under a scenario in which phosphorus loading to the lake is reduced by 40% from 2015 levels. They estimate the present value of climate benefits from the CH_4 reductions (if maintained through 2050) to be \$3.1 billion. The study demonstrates that the climate benefits of nutrient management policies can be substantial, but the analysis included only a single GHG and a single waterbody. While studies have found that CH_4 emissions from lakes and reservoirs contribute more to the radiative forcing of the atmosphere than emissions of CO_2 and N_2O (Deemer et al. 2016), the global warming potential of N_2O is an order of magnitude greater than that of CH_4 (Forster et al. 2021) and is thus also worthy of study, particularly in agricultural regions where high nitrogen availability can stimulate N_2O emissions. The present study expands understanding of the climate benefits of water quality policies by evaluating changes in all three primary GHGs (CH_4 , CO_2 , and N_2O) and is placed in the context of policies that affect the vast majority of surface waters in the U.S.

In this study, we focus on the climate benefits of a nutrient management program in the Chesapeake Bay Watershed (Fig. 1). Under the Clean Water Act, the U.S. Environmental Protection Agency

(EPA) initiated the Chesapeake Bay Total Maximum Daily Load (TMDL) in 2010. The TMDL is designed to achieve water quality standards by reducing annual loadings of nitrogen and phosphorus by 25% and 24%, respectively, from the base year of 2009. Under the TMDL, the six mid-Atlantic states and the District of Columbia that contain the Chesapeake Bay Watershed were required to develop nutrient and sediment management plans and to ensure that all control measures are fully implemented by 2025. A linked modeling system of the airshed, watershed, and estuary was developed to ensure the plans would meet the TMDL targets and to estimate the economic benefits of improving water quality in the Bay. We extend that set of models to estimate the corresponding annual reductions in CO₂, CH₄, and N₂O emissions from Chesapeake Bay Watershed lakes and reservoirs and apply a set of social cost of GHG (SC-GHG) estimates recently developed by EPA (U.S. EPA 2023d) to monetize the climate benefits associated with these emissions reductions. Lastly, we extrapolate our results to illustrate the magnitude of climate benefits from a similar nutrient management program in the Mississippi-Atchafalaya River Basin (Fig. 1), the largest U.S. watershed containing nearly 200,000 waterbodies.

2 MATERIALS AND METHODS

2.1 Modeling approach

We calculate the climate benefits of the Chesapeake Bay TMDL by modeling the physical and biogeochemical processes (Fig. 2) with and without the policy (TMDL and Baseline scenarios, respectively), then calculate the difference in GHG emissions between the scenarios. The Baseline scenario is developed from model inputs representing 2009 conditions, and the TMDL scenario is modeled for 2025, the first year of full implementation of management practices under the policy.

The modeling begins with mapping nutrient loadings to waterways from agricultural and industrial sources, wastewater treatment, stormwater runoff, and air deposition (Chesapeake Bay Watershed Model). This model is linked to a nutrient transport and retention model (the “Northeast Lakes Model”) that provides total phosphorus (TP) and chlorophyll-a concentrations (an indicator of algae abundance) in each of the more than 4,000 lakes and reservoirs in the watershed. Output of the Northeast Lakes Model is linked to our “Greenhouse Gas Emissions Model” to estimate annual emissions of CO₂, CH₄, and N₂O from each waterbody, accounting for seasonal variation in GHG emission rates. Time step and performance of each model are summarized in Table 1. Finally, we monetize climate benefits by applying year- and gas-specific SC-GHG values to the estimated emissions reductions of each GHG in each year.

2.2 Nutrient loadings to the stream network

We estimate mean-annual nutrient loadings to the stream network under the Baseline and TMDL scenarios using Phase 5.3 of the Chesapeake Bay Watershed Model which simulates the transport and fate of nitrogen, phosphorus, and sediment from 309 land segments to each of 1,069 river segments (U.S. EPA 2010). Land uses within the model include 13 types of cropland, two types of woodland, three types of pasture, four types of urban land, and other special land uses such as surface mining and construction (Shenk and Linker 2013). Nutrient sources include manure, fertilizer, wastewater discharges, septic system loads, and atmospheric deposition. EPA's Chesapeake Bay Program Office developed scenarios to simulate baseline and TMDL conditions. The key model inputs for the TMDL scenario are based on the surrounding jurisdictions' Watershed Implementation Plans (WIPs) (U.S. EPA 2023a). WIPs detail how each jurisdiction will achieve the TMDL allocations by 2025, including target reductions for each pollutant source sector, the management practices that will provide those reductions, and where those practices will be implemented.

2.3 Lake nutrient and chlorophyll-a concentrations

The Northeast Lakes Model (Milstead et al. 2013) is used to predict total nitrogen (TN) and TP concentrations in each lake and reservoir under the Baseline and TMDL scenarios. The model estimates TN and TP concentrations from nutrients delivered to waterbodies from the upstream river network, lake hydraulic residence time, and mean depth. Nutrient loadings to the waterbodies are taken from the Chesapeake Bay Watershed Model outputs, hydraulic residence time is calculated as the ratio of volume to flow, and mean depth is calculated as the ratio of volume to surface area. Chlorophyll-a concentration in each waterbody is calculated using a multiple linear regression model with TP and TN as predictors parameterized using waterbody-specific data collected during EPA's 2007 National Lakes Assessment (U.S. EPA 2016).

2.4 Greenhouse gas emissions

Lake surface area data and the TP and chlorophyll-a projections from the Northeast Lakes Model are used as inputs to a set of greenhouse gas emissions models (DeSontro et al. 2018) (Table 1) to predict areal GHG emission rates for each lake under the Baseline and TMDL scenarios. The GHG model predictions reflect the model training data, which were emission rates measured primarily during the summer, and likely do not incorporate seasonal variability. Seasonality in CO₂ and N₂O emission rates is not well

understood, with some studies reporting higher rates during the summer (Beaulieu et al. 2010; Natchimuthu et al. 2017) and others during the winter (Kortelainen et al. 2020; Xiao et al. 2020; Sun et al. 2021). We assume that CO₂ and N₂O emission rates predicted by the Delontro et al. (2018) models are applicable throughout the year.

Methane production rates exhibit strong seasonality with warm season emission rates far exceeding cold season rates (Aben et al. 2017). We assume a mean cold season (November 1 – April 1) CH₄ production rate of 10.56 mg CH₄ m⁻² d⁻¹ based on the results of an eddy covariance continuous monitoring study at a reservoir in southwestern Ohio (Waldo et al. 2021a). During period of ice-cover we assume this CH₄ accumulates under the ice, along with CO₂ and N₂O.

Accumulated CO₂ and N₂O rapidly vent to the atmosphere during ice out, but a fraction of the accumulated CH₄ is converted to CO₂ or microbial biomass by methanotrophy, a type of microbial metabolism. We model the fate of CH₄ trapped under the ice by 1) assigning annual ice-cover duration for each waterbody as the ten year mean from the ERA5-Land database for 2010–2020 (Copernicus Climate Change Service (CS3) 2021), 2) assuming the fraction of accumulated CH₄ that is subject to methanotrophy ranges from 0.01-0.6 (Michmerhuizen et al. 1996; Kankaala et al. 2006), and 2) assuming the fraction of CH₄ converted to microbial biomass via methanotrophy (i.e. methanotrophic bacterial growth efficiency) ranges from 0.05–0.8 (Bastviken et al. 2003). The TMDL effect on winter CH₄ emission rates was assumed identical to the proportional change in each lake’s warm weather CH₄ emission rate under the TMDL. See simulation section below for uncertainty details.

2.5 Climate benefits of greenhouse gas emission reductions

We estimate the climate benefits of the GHG reductions from the Chesapeake Bay TMDL using estimates of the SC-GHG, specifically the social cost of CO₂ (SC-CO₂), social cost of CH₄ (SC-CH₄) and social cost of N₂O (SC-N₂O). These gas- and year-specific values combine climate science and economics to put the effects of climate change into monetary terms to help policymakers and the public understand the societal consequences of actions that would increase or decrease GHG emissions. The SC-GHG is the monetary value of the net harm to society associated with adding a small amount of that GHG to the atmosphere in a given year. The SC-GHG estimates used in this analysis were taken from a recent update (U.S. EPA 2023d) that incorporates scientific advances in climate science and economics and provides explicit representation of many sources of uncertainty, as recommended by the National Academies of Sciences,

Engineering, and Medicine (NASEM 2017). The updated values are larger than estimates used in earlier EPA analyses and in the study of the climate benefits of reducing nutrient pollution by Downing et al. (2021). The new estimates draw on the latest research and expertise from the scientific disciplines relevant to the SC-GHG estimation process. Socioeconomic inputs were taken from probabilistic projections for population, income, and GHG emissions (Rennert et al. 2022b). Climate trajectories resulting from the GHG emissions were developed with the Finite Amplitude Impulse Response (FaIR) model (Millar et al. 2017; Smith et al. 2018), a widely used simple Earth system model which captures the relationships between GHG emissions, atmospheric GHG concentrations, and global mean surface temperature change. The socioeconomic projections and outputs of the climate modeling were used as inputs to damage functions that estimate monetized future damages from temperature change (Rode et al. 2021; Carleton et al. 2022; Climate Impact Lab (CIL) 2022; Rennert et al. 2022a; Rennert et al. 2022b). Finally, the stream of future climate damages was discounted back to the year of emissions using a set of dynamic discount rates that captures the role of uncertainty in the discount rate following the latest empirical evidence on interest rate uncertainty. The discounting parameters were calibrated to a near-term target discount rate of 2 percent following Newell et al. (2022), as applied in Rennert et al. (2022a; 2022b). The estimation process is performed in 10-year intervals from 2020 through 2080, and linearly interpolated between the decadal timestep to recover annual estimates for the SC-GHGs. EPA (2023d) provides a full discussion of the methodology, and the annual SC-GHG estimates are provided in Extended Data Table 2.

2.6 Uncertainty

Uncertainty in annual GHG emissions were assessed by running 10,000 iterations of the GHG model using a monte carlo simulation. The simulation incorporated uncertainty in predictions from the Greenhouse Gas model but not the Chesapeake Bay or Northeast Lakes models (see Table 1 for model performance).

A unique set of GHG model coefficients were drawn for each iteration of the simulation based on the distribution of the model coefficients (e.g. mean and standard error) and their covariance matrix (Krinsky and Robb 1986). Cold season (November 1 – April 1) CH₄ emission rates were drawn from a normal distribution with a mean of 10.56 and a standard deviation of 11.52 mg CH₄ m⁻² d⁻¹ (Waldo et al. 2021a). Values drawn from the distribution were truncated at 0 and 4,000 mg CH₄ m⁻² d⁻¹ to ensure environmentally realistic values. The fraction of accumulated CH₄ that is subject to methanotrophy was simulated as a uniformly distributed variable ranging from 0.01 to 0.6 (Michmerhuizen et al. 1996;

Kankaala et al. 2006). The fraction of CH₄ converted to microbial biomass via methanotrophy was simulated as a uniformly distributed variable ranging from 0.05 – 0.8 (Bastviken et al. 2003). Areal emission rates were scaled by waterbody surface area and results are reported as the mean and 95% confidence interval of the 10,000 iterations.

The SC-GHG estimates reflect a quantitative consideration of many sources of uncertainty. Specifically, the SC-GHG estimates are presented as certainty-equivalent values—defined in economics as the guaranteed outcome that an individual would consider equivalent to the value of a risky outcome. The certainty-equivalent SC-GHG is the mean of the discounted marginal damages for each emissions year and GHG. The willingness to pay for reduced uncertainty, the risk premium, is capitalized into the SC-GHG when collapsed to the mean. Each distribution of marginal damages consists of 10,000 estimates based on draws from the distributions of uncertain parameters in the socioeconomic, climate and damage function components of the modeling. See U.S. EPA (2023d) for more discussion of the key modeled uncertainties and their sources.

Note that the simplifying assumption of constant emission reductions beyond 2025 likely leads to an underestimate of the climate benefits of the Chesapeake TMDL. Under the *Baseline* scenario, annual loadings are held constant at 2009 levels, while the TMDL specifies a loadings cap. Since development in the watershed is expected to increase over time (Ator et al. 2020), meeting the TMDL will likely require larger reductions in pollutant loadings in future years (and hence larger emissions reductions) than what we have estimated in this analysis.

2.7 Statistical analysis of spatial patterns

Spatial trends in phosphorus, chlorophyll-a, and GHG emission rates reductions across lakes were quantified via Trend Surface analysis (Cressie 1993) where each variable was modeled as a function of the longitude, latitude, and their interaction, of the centroid of each lake. Variable significance was assessed at the 0.05 level and the final models were used to predict each response variable throughout a grid spanning the spatial domain of the watershed (Extended Data Fig.1). Trend Surface Analysis was also used to assess spatial patterns in land-cover within the Chesapeake Bay Watershed.

2.8 Extrapolation to Mississippi River Basin

A root cause of the largest coastal eutrophication zone in the U.S. is nutrient loadings from the Mississippi-Atchafalaya River Basin (MARB)—where the fertile soil supports the largest corn and grain production in

the world (USDA 2023). EPA’s Mississippi River/Gulf of America Hypoxia Task Force has adopted a goal of reducing the average annual size of the Gulf’s hypoxic zone to no more than 5,000 square kilometers by the year 2035—an ambitious goal which could require reductions in TN and TP loading to the Gulf of 48% (Greene et al. 2009; Fennel and Laurent 2018; U.S. EPA 2025). The Taskforce action plan includes strategies mirroring those underlying the TMDLs in the Chesapeake Bay (such as an interim target of a 20% reduction of nitrogen and phosphorus loading by 2025). Given these similarities, we extrapolate our findings to illustrate the climate benefits from a nutrient management policy in the MARB comparable to the Chesapeake TMDL. This extension assumes that nutrient management in the MARB will reduce total nitrogen and phosphorus comparable to the Chesapeake Bay TMDLs and, thus, areal CH₄ emission rates (CH₄ tons per hectare of water body surface area) by the same proportion as the modeled reductions for Chesapeake Bay waterbodies. Baseline areal GHG emission rates for MARB lakes and reservoirs in 2021 are taken from the U.S. Inventory of Greenhouse Gas Emissions and Sinks (U.S. EPA 2023b). We restrict the extrapolation to CH₄ because that is the only GHG reported for waterbodies in the inventory. We then apply the mean percent reductions in areal CH₄ emission rates predicted in the Chesapeake waterbodies to the MARB waterbodies by climate zone: the boreal MARB waterbodies are mapped to the cool temperature waterbodies in the Chesapeake Watershed and all other MARB waterbodies are mapped to Chesapeake waterbodies in the warm temperate moist climate zone.

3 RESULTS AND DISCUSSION

3.1 Nutrient, chlorophyll-a, and areal greenhouse gas emission rate reductions

The linked Chesapeake Bay Watershed and Northeast Lakes models predict that implementation of the TMDL policy will reduce TP and chlorophyll-a in 98.2% of the lakes and reservoirs in the watershed. The cumulative distribution plots for chlorophyll-a and TP under the TMDL scenario mirror those for the Baseline scenario, indicating that the TMDL policy results in lower chlorophyll-a and TP across the full range of concentrations encountered in the watershed (Fig. 3A, 3C). The TMDL policy is predicted to reduce TP and chlorophyll-a concentrations by 21% from baseline (average of 18 and 7 ug/L, respectively), across the 4,221 waterbodies (Extended Data Table 1). The greatest reductions in TP and chlorophyll-a are predicted to occur in the southern portion of the watershed (Fig. 3B, D, Extended Data Fig. 1A, B, trend surface analysis $p < 2.2e-16$), likely reflecting an uneven distribution of land cover and management practices under the TMDL scenario. Croplands and urban land cover exhibit a strong latitudinal gradient in the watershed ($p < 2.2e-16$) with northern portions primarily forested and southern portions developed

for agriculture and settlements. Best management practices tend to be sited in areas of high nutrient loading, thus few have been implemented in the northern portion of the basin; nutrient load reductions are not expected to exceed 5 % in that region (Sekellick et al. 2019).

The GHG models translate the decreases in TP and chlorophyll-a to changes in GHG emissions from the waterbodies. Areal CH_4 , CO_2 , and N_2O emission rates (mass GHG/unit area water surface/unit time) are estimated to decrease by an average of $6.6 \text{ g CH}_4 \text{ m}^{-2} \text{ year}^{-1}$ (17% reduction), of $42 \text{ g CO}_2 \text{ m}^{-2} \text{ y}^{-1}$ (4.9% reduction), and $0.006 \text{ g N}_2\text{O m}^{-2} \text{ y}^{-1}$ (7.3 % reduction), respectively, under the TMDL policy, relative to the Baseline scenario (Fig. 4). For all three gases, the greatest emission rate reductions are predicted to occur in the southern portion of the basin (Fig. 4, Extended Data Fig. 1C – E, trend surface analysis $p < 2.2\text{e-}16$), consistent with where the greatest reductions in chlorophyll-a and TP are predicted to occur (Fig. 3B, D). Gas-specific factors also play a role in the spatial distribution and relative magnitude of the reductions. For methane, the spatial pattern in emission rate reductions is enhanced by the latitudinal pattern in ice-cover duration across the watershed (trend surface analysis $p < 2.2\text{e-}16$). Lakes in the northern portion of the watershed experience up to 100 days of ice cover per year (annual average 2010-2020) whereas those in the southern portion are ice covered for 5 or fewer days per year (Extended Data Fig. 2). The amount of CH_4 that is subject to methanotrophy during ice out is proportional to ice-cover duration, therefore the northern lakes lose a greater proportion of their CH_4 to methanotrophy than lakes in the southern portion of the watershed. Thus, for any given waterbody size and TP concentration, northern lakes will emit less CH_4 per year than their southern counterparts. By extension, identical proportional reductions in TP concentration will result in greater emission reductions in southern versus northern lakes. Our modeling is among the first to account for seasonality in CH_4 emissions, CH_4 storage under ice, and methanotrophy during ice-out. Methane emissions during the cold-weather season, including emissions during ice-out, constituted only 8.7 % (5.9 – 12.0% 95% CI) of annual emissions under Baseline conditions and 8.0% (5.6 – 10.8 % 95% CI) under the TMDL scenario, which is consistent with numerous reports that biological CH_4 production is strongly temperature dependent (Yvon-Durocher et al. 2014; Waldo et al. 2021a). Our modeling predicts reductions in cold-season CH_4 emissions constitute only 7% (146 metric tons) of annual avoided emissions (2041 metric tons) from the policy.

As with CH_4 , areal CO_2 emission rate reductions are predicted to be greatest in the southern portion of the watershed where nutrient load reductions are expected to be largest. CO_2 emission rate reductions are predicted to be inversely related to waterbody size. The relationship between CO_2 emission rate and TP is positive in smaller lakes ($< \text{approximately } 12 \text{ km}^2$), negative in larger lakes, and nearly non-

existent in mid-size waterbodies (DeISontro et al. 2018). Hence, reductions in TP yielded the largest reductions in areal CO₂ emission rates in smaller lakes (Extended Data Fig. 3). On the other hand, the models predict that the policy's TP reductions will lead to an increase CO₂ emission rates in the four lakes that are larger than 12 km², but the magnitude of change is relatively small (on average 0.33 g CO₂ m⁻² y⁻¹). Our modeling does not include seasonality in CO₂ emissions due to a lack of data and this is an important topic for future research.

Reductions in N₂O emission rates are also greatest in the southern portion of the watershed, but the pattern is not as pronounced as it is for CH₄ and CO₂, possibly because the relationship between areal N₂O emission rates and chlorophyll-a is confounded by lake size. The model predicts that areal N₂O emission rates will increase with both chlorophyll-a and lake size, thus the spatial distribution of reductions in areal N₂O emission rates will be driven by the spatial patterns in lake size and chlorophyll-a reductions.

3.2 Annual greenhouse gas emission reductions

Taken together, the linked Chesapeake Bay Watershed, Northeast Lakes, and Greenhouse Gas models (Fig. 2) predict that the TMDL policy, once fully implemented, will reduce annual GHG emissions from waterbodies in the Chesapeake Bay Watershed by 16.4, 3.2, and 5.9% for CH₄ (95% confidence interval of 1199-3158 metric tons), CO₂ (7871, 6646–9123 metric tons), and N₂O (2.6 (1.2-4.2) metric tons), respectively (Table 2). These results reflect the net effect of the differences in the relationship between areal emission rates (mass of GHG/unit area/unit time) and environmental drivers across the three gases. Carbon dioxide and N₂O areal emission rates depend on the waterbody surface area, whereas CH₄ emission rates are independent of waterbody size. The distribution of waterbody sizes in the watershed is characterized by numerous smaller waterbodies and relatively few larger systems (Extended Data Fig. 4). For example, waterbodies with surface areas greater than 1 km² constitute less than 2% of the 4,221 waterbodies in the watershed, but they make up 43% of the total waterbody surface area (210 km²) (Extended Data Fig. 4). Despite their expansive cumulative surface area, these waterbodies contribute little to the reduction in CO₂ emissions under the TMDL policy (Fig. 5) because the GHG model predicts very low areal CO₂ emission rates in large lakes. By contrast, areal N₂O emission rates are predicted to increase with lake size and 37% of the reduction in N₂O emissions can be attributed to the 54 waterbodies larger than 1 km². For methane, the GHG model predicts no relationship with lake size; 29% of the reduction in CH₄ emissions occurred in large lakes. Thus, the effect of a nutrient management

policy on CO₂ and N₂O emissions from waterbodies is strongly dependent on the distribution of lake sizes within the target watershed, whereas the policy's effect on CH₄ emissions is less so.

3.3 Monetized climate benefits

Applying the SC-CO₂, SC-CH₄, and SC-N₂O estimates for 2025 to the emission reductions expected under full implementation of the TMDL scenario in 2025 yields climate benefits of over \$6 million in 2025. SC-GHG estimates increase over time because emissions further in the future produce larger incremental damages as physical and economic systems become more stressed and because incomes will continue to grow in the future. Thus, holding the emission reductions constant, annual monetized climate benefits are expected to increase beyond 2025. By 2075, for example, the SC-CO₂, SC-CH₄, and SC-N₂O values (based on a 2% near-term discount rate) rise to \$391 per metric ton CO₂, \$6,355 per metric ton CH₄, and \$123,926 per metric ton N₂O (2020USD), respectively. Applying these 2075 SC-GHG values to the 2025 emission reduction estimates above yields monetized climate benefits of about \$16 million by the year 2075 (Table 2). The net present value of the monetized climate benefits over the first fifty years of the program (2025-2075) are estimated to be \$333 million (2020USD, 2% discount rate).

The majority of the climate benefit of the policy is derived from avoided CH₄ emissions, followed by CO₂, with N₂O making a minor contribution (Table 2). While SC-N₂O is 27 and 255 -fold greater than that of CH₄ and CO₂, respectively (Table 2), the mass of avoided N₂O emissions is an order of magnitude less than that of CH₄ and CO₂. Areal N₂O emission rates tend to be much lower than that of CH₄ and CO₂ (on a per mass basis) (DelSontro et al. 2018) but have been increasing over the last century (Wang et al. 2023) and may become a more important target for emission reductions in the future.

The monetized climate benefits from CH₄ reductions are considerably smaller than those estimated by Downing et al. (2021) for reducing nutrient pollution to Lake Erie, but that is to be expected given the vastly greater surface area of Lake Erie (25,667 km²) relative to that of the Chesapeake Bay waterbodies (491 km²). A more useful comparison is to place our estimates in the context of other estimated benefits of the Chesapeake Bay TMDL. The most common monetized category of benefits in BCAs of Clean Water Act regulations is recreational fishing. Massey et al.(2017) estimate monetary benefits from increased recreational catch in Chesapeake Bay to range from \$6 to \$70 million annually. Moore and Griffiths (2018) estimate the benefits to the commercial fishing industry to be \$16 million annually. Finally, Klemick et al. (2016) find that the impact on property values to all homes near the Bay to range from \$32 to \$60 million per year when annualized using a 7% discount rate. The climate benefits of the Chesapeake Bay TMDL are similar in magnitude to other commonly quantified categories of water quality benefits, but are likely

conservative because our analysis does not include potential reductions in GHG emissions from streams (Smith et al. 2017), rivers (Cole and Caraco 2001), Chesapeake Bay (Ji et al. 2018) or agricultural soils in the watershed (Gasper et al. 2012; Fisher et al. 2018; Weng et al. 2023).

Note that the simplifying assumption of constant emission reductions beyond 2025 also likely leads to an underestimate of the climate benefits of the Chesapeake TMDL. Under the *Baseline* scenario, annual pollutant loadings are held constant at 2009 levels, while the TMDL specifies a loadings cap. Since development in the watershed is expected to increase over time (Ator et al. 2020), meeting the TMDL will likely require larger reductions in pollutant loadings in future years (and hence larger emissions reductions) than what we have estimated in this analysis.

3.4 Mississippi-Atchafalaya River Basin extrapolation

Our integrated analysis focuses on a major nutrient management program in the U.S., but similar programs are being implemented in watersheds throughout the country (U.S. EPA 2023c) and the cumulative climate benefits of these policies could be substantial. For example, the EPA's Mississippi River/Gulf of America Hypoxia Task Force targets reducing nitrogen and phosphorus loading from the Mississippi-Atchafalaya River Basin (MARB, Fig. 1) to reduce the average annual extent of the Gulf's hypoxic zone to less than 5,000 square kilometers by the year 2035. We estimate the climate benefits of the nutrient management policy under the assumption it will reduce areal CH₄ emission rates from MARB waterbodies by the same proportion as our modeled reductions for Chesapeake Bay waterbodies (see Methods). This extrapolation does not account for the spatial distribution of management practices in the MARB or differences in ice-cover duration and chlorophyll-a between the MARB and Chesapeake Bay Watershed and is therefore more uncertain than our Chesapeake Bay analysis. We estimate the policy will reduce annual CH₄ emissions from MARB waterbodies by 103,404 metric tons. Applying the 2035 SC-CH₄ of \$2,842/mt CH₄ (based on a 2% near-term discount rate) to these reductions, we estimate the climate benefits to be \$290 million (2020USD) in 2035. This is a non-trivial benefit when compared to estimated abatement costs (e.g., \$1.4 - 2.7 billion/yr) for policies achieving nitrogen and phosphorus load reductions of about 19 - 36% (Rabotyagov et al. 2010; 2014). The net present value of the monetized climate benefits from achieving these CH₄ emission reductions for forty years (2035-2075) are then \$10.6 billion (2020USD, 2% discount rate). This is likely a conservative estimate of the climate benefits because it does not account for the accompanying CO₂ and N₂O reductions, and like the Chesapeake Bay study, overlooks many other climate and non-climate (Parthum and Ando 2020) benefits of the policy (e.g. reduced GHG emissions from streams).

4 CONCLUSIONS

The relationship between GHG emission rates and nutrient loading has been demonstrated from global (Deemer et al. 2016; DelSontro et al. 2018) to local scales (Liao et al. 2023) and is generalizable across a wide range of systems. In this study we applied the relationship to a nutrient management policy in the Chesapeake Bay watershed and found that the policy will reduce emissions of CO₂, CH₄, and N₂O from waterbodies in the watershed and generate climate benefits that are comparable in magnitude to other commonly estimated benefits (e.g., recreational, and commercial fishing). While we take a case study approach, nutrient management plans are ubiquitous throughout the U.S. Between 2000 and 2023 the EPA funded 21,414 nonpoint source watershed projects across all 50 states (U.S. EPA 2023c) and administered 33,820 nutrient TMDLs in 2011 (U.S. EPA 2011). Furthermore, excessive nutrient loading of surface waters extends far beyond the U.S. with an estimated 1.15 million km² of coastal waters at risk of eutrophication worldwide (Maúre et al. 2021) causing many nations to adopt nutrient management policies. The climate benefits of nutrient management could be substantial at regional, national, and global scales.

We demonstrate that the climate benefits of nutrient management programs can be sufficiently large to merit their inclusion in the BCA of water quality policies. These benefits may also be relevant to other types of environmental policies including carbon-credit markets where, for example, management practices designed to promote soil carbon storage (e.g. no-till farming) may also improve water quality and result in reduced GHG emissions from waterbodies. These examples illustrate that a comprehensive consideration of the climate benefits of policies may further incentivize sustainable practices, leading to improved environmental outcomes.

Data and Code Availability

All data and replication code supporting this study's findings are hosted on a public GitHub repository and Zenodo that will be published alongside the manuscript.

Conflict of Interests

The authors have no competing interests to report.

Author Contributions

Jake J. Beaulieu: Methodology; software; formal analysis; data curation; writing - original draft; writing - review & editing; visualization. **Elizabeth Kopits:** Methodology; writing - original draft; writing - review & editing; visualization. **Chris C. Moore:** Methodology; visualization; writing - original draft; writing - review & editing; visualization. **Bryan M. Parthum:** Methodology; software; formal analysis; data curation; writing - original draft; writing - review & editing; visualization.

Acknowledgements

The authors would like to thank participants of the Social Cost of Water Pollution Workshop and the Cornell Atkinson Center for Sustainability for helpful feedback and conversation. The views expressed in this article are those of the authors and do not necessarily represent the views or policies of the U.S. Environmental Protection Agency.

Main Display Elements

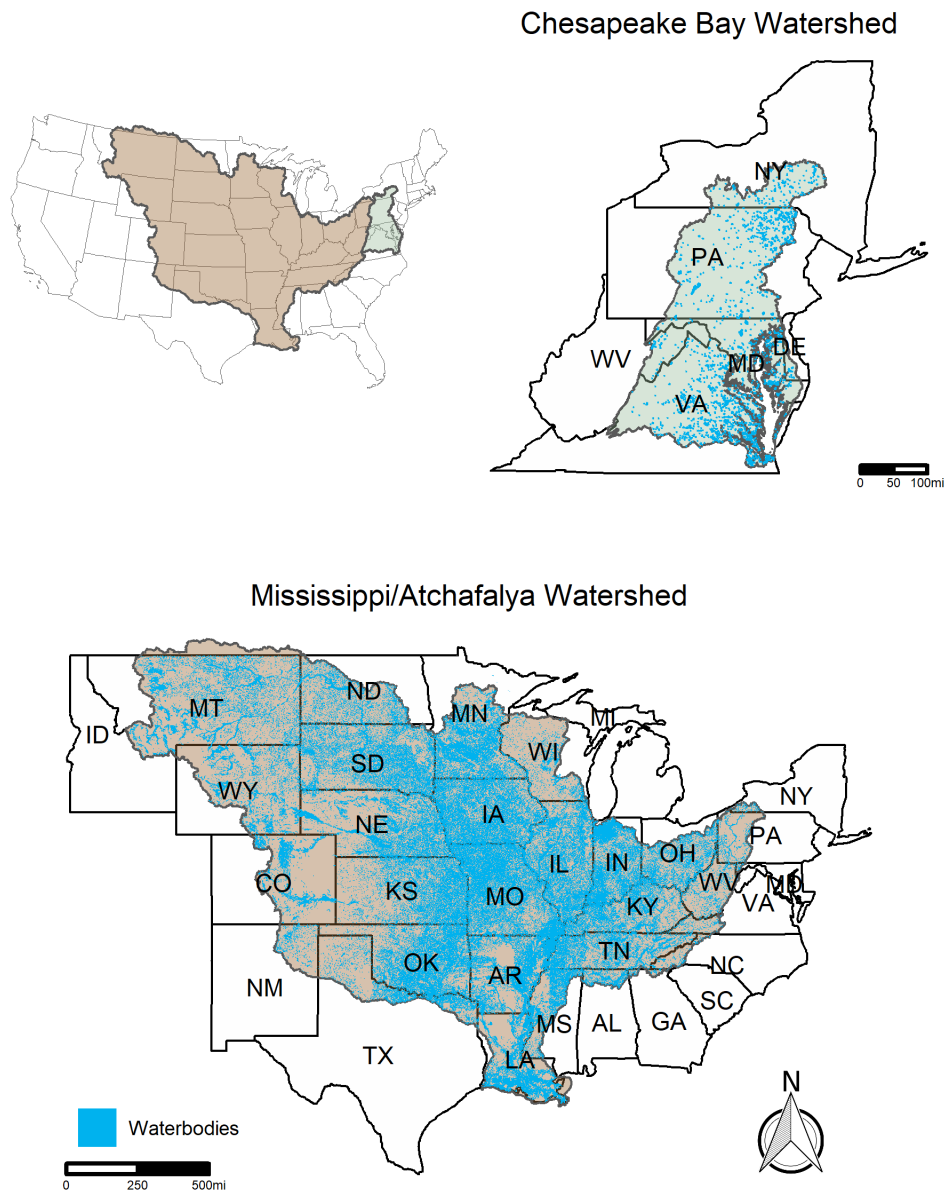


Fig. 1 | Lakes and reservoirs in the Chesapeake Bay and Mississippi-Atchafalaya Watersheds. The Chesapeake Bay Watershed (green region) is in the Northeastern region of the United States, spanning six states and the District of Columbia. Within the watershed, there are roughly 4,200 waterbodies that are affected by the Total Maximum Daily Load policy for nitrogen and phosphorus. The Mississippi-Atchafalaya Watershed (brown region) drains parts of 31 states and covers 3.2 million km².

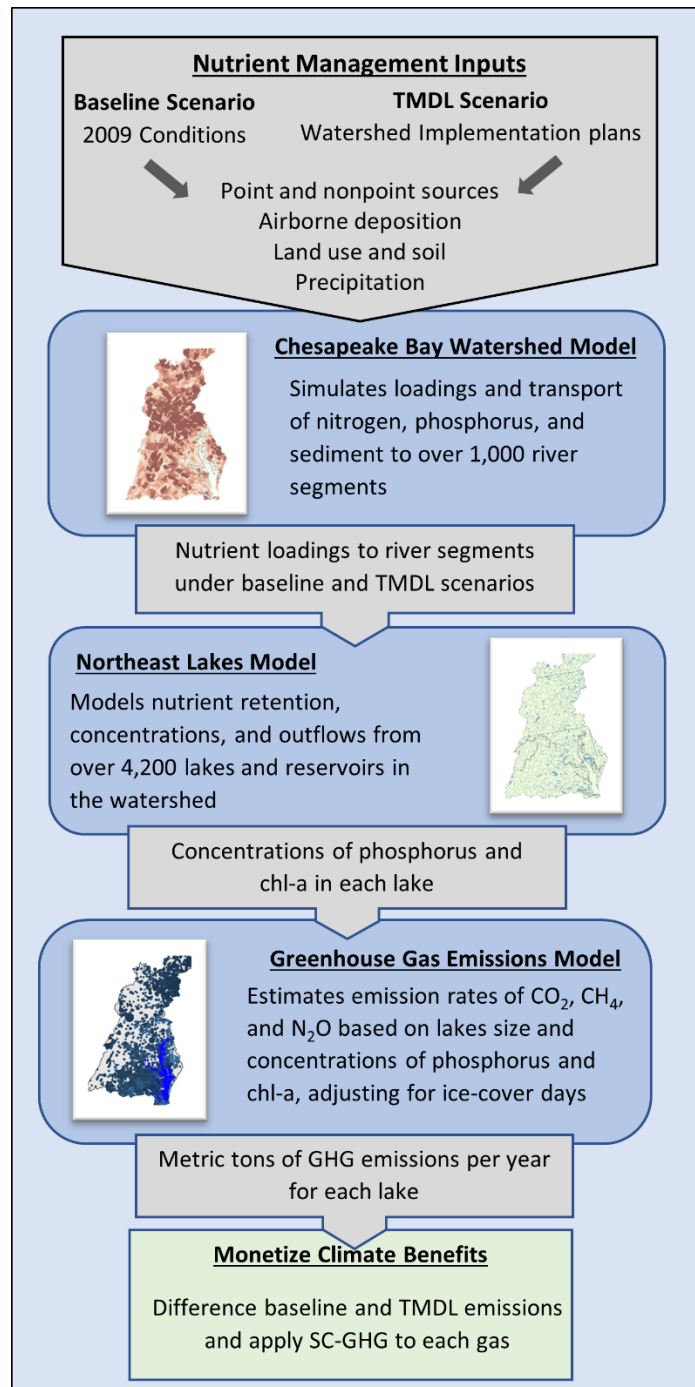


Fig. 2 | Modeling pathway linking nutrient management programs to climate benefit endpoints. Framework for estimating climate benefits from nutrient management policies such as the Chesapeake Bay TMDL.

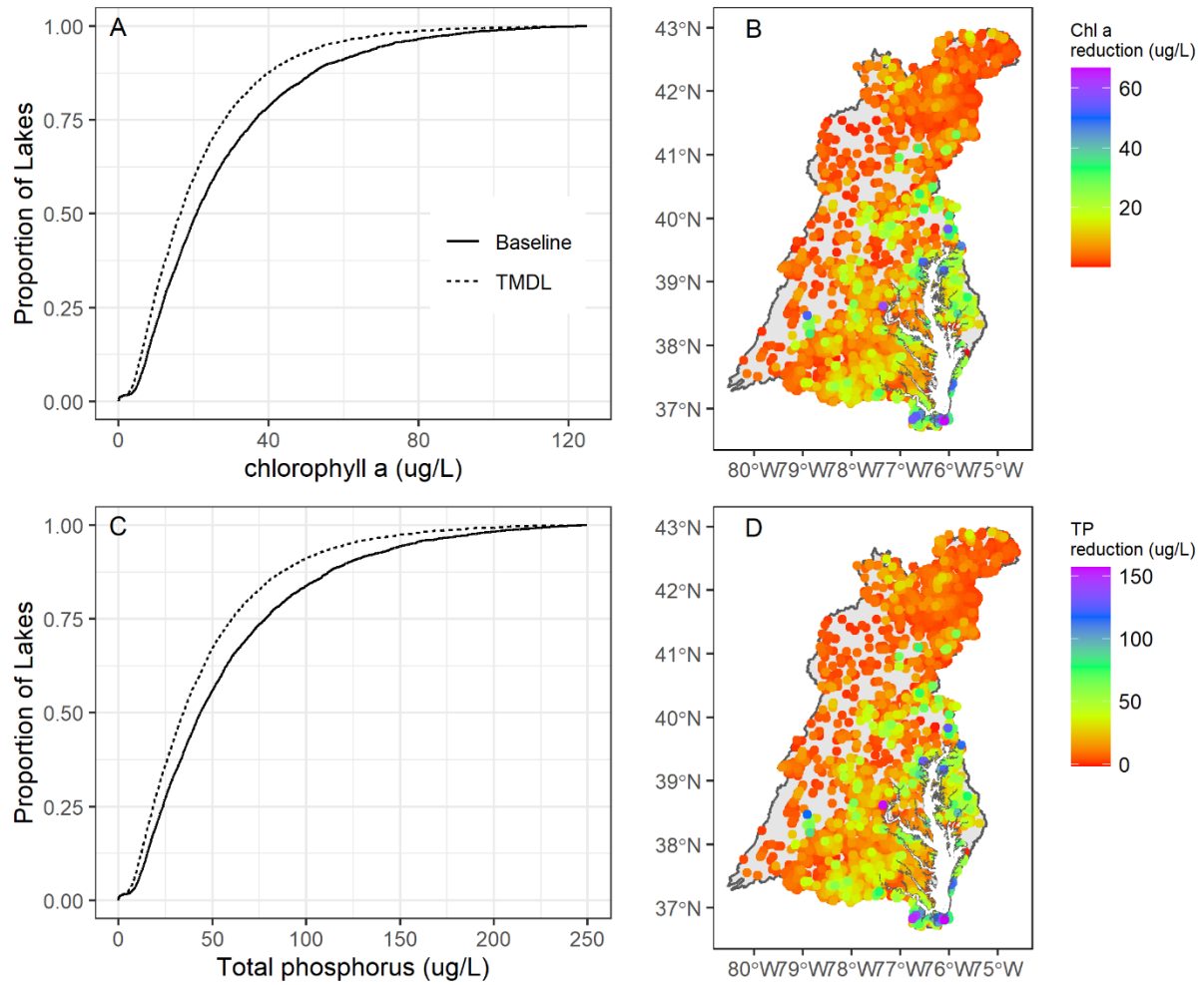


Fig. 3 | Heterogeneity in nutrient reductions from the Chesapeake Bay TMDL. The modeled effect of the Chesapeake Bay TMDL on lake chlorophyll-a and total phosphorus (TP) varies across waterbodies and regions. This can be seen using cumulative distribution plots of chlorophyll-a (Panel A) and TP (Panel C) concentrations under Baseline and TMDL scenarios and the spatial distribution of these reductions in chlorophyll-a (Panel B) and total phosphorus (Panel D) after implementation of the TMDL policy. Less than 2% of the lakes and reservoirs in the watershed had higher chlorophyll-a and total phosphorus under the TMDL policy.

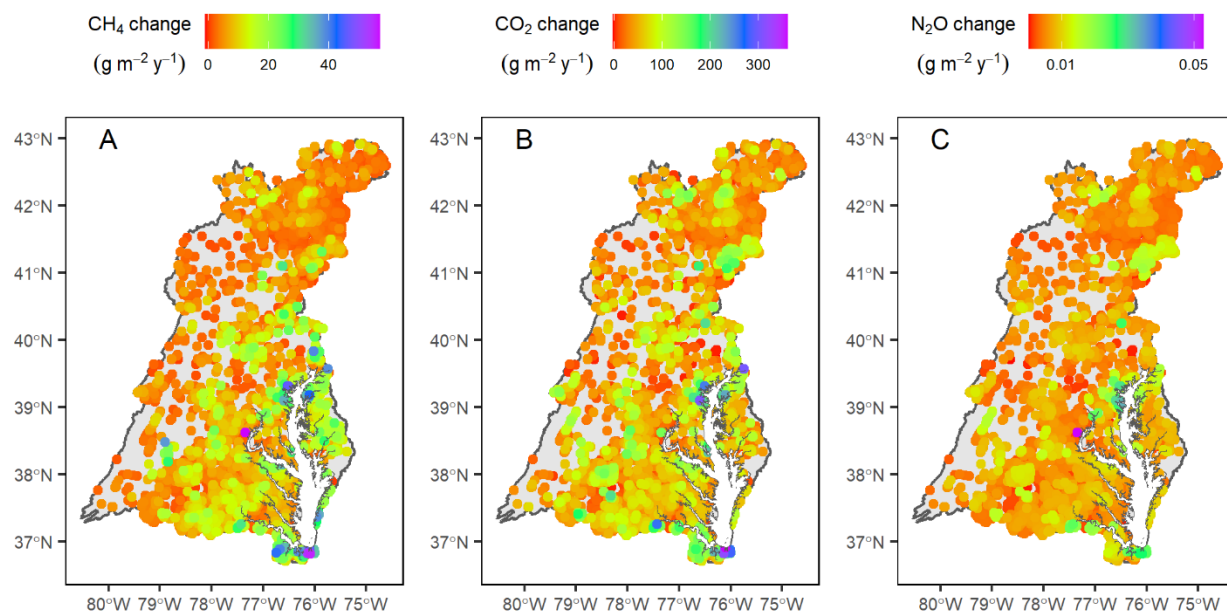


Fig. 4 | Waterbody-specific reductions in greenhouse gas emissions. The spatial heterogeneity in greenhouse gas reductions attributable to The Chesapeake Bay TMDL in A) methane (CH₄), B) carbon dioxide (CO₂), and C) nitrous oxide (N₂O) areal emission rates under the TMDL policy, relative to the Baseline scenario.

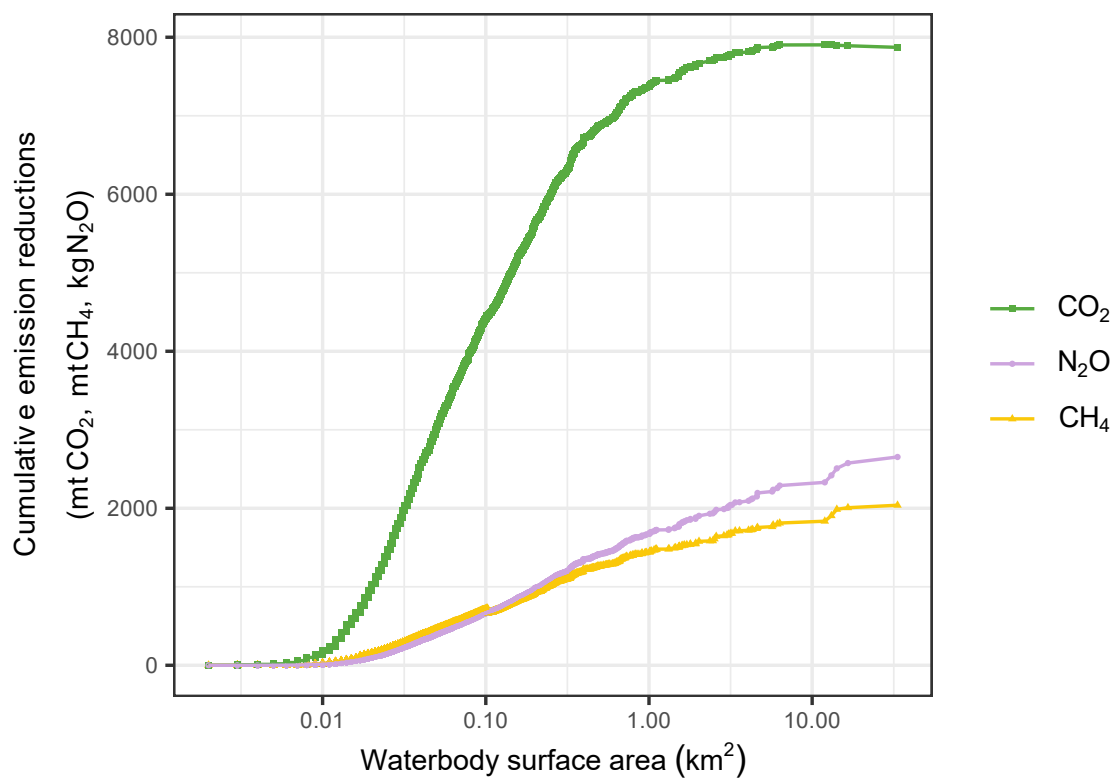


Fig. 5 | Cumulative reductions in greenhouse gas emissions plotted against lake size. While large lakes contributed relatively little to reductions in carbon dioxide (CO₂) emissions, reductions in nitrous oxide (N₂O) and methane (CH₄) are relatively linear across waterbody surface area.

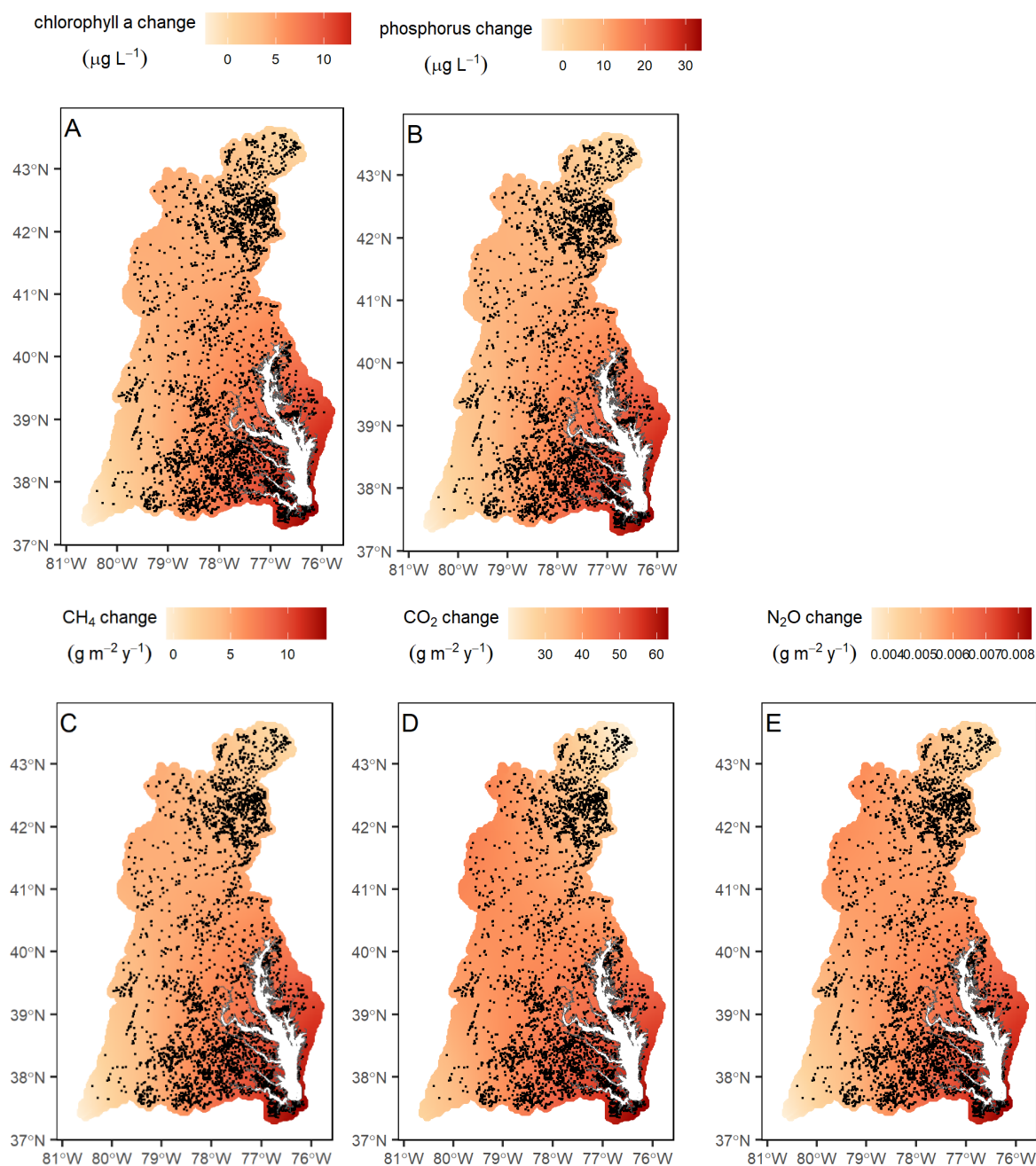
Model	Parameter	Prediction scale		Model performance	
		Temporal	Spatial	r ²	RMSE
Chesapeake Bay Watershed U.S. EPA 2010	TP	annual	stream reach	0.91	0.65
	TN			0.97	0.35
Northeast Lakes Milstead et al. 2013	TP	summer	lake	0.54	0.36
	Chl a			0.78	0.25
Greenhouse Gas Delsontro et al. 2018	CO ₂	seasonal	lake	0.11	0.33
	N ₂ O			0.09	0.20
	CH ₄			0.38	0.55

Table 1 | Prediction scale and performance of models used to estimate climate benefits of nutrient management.

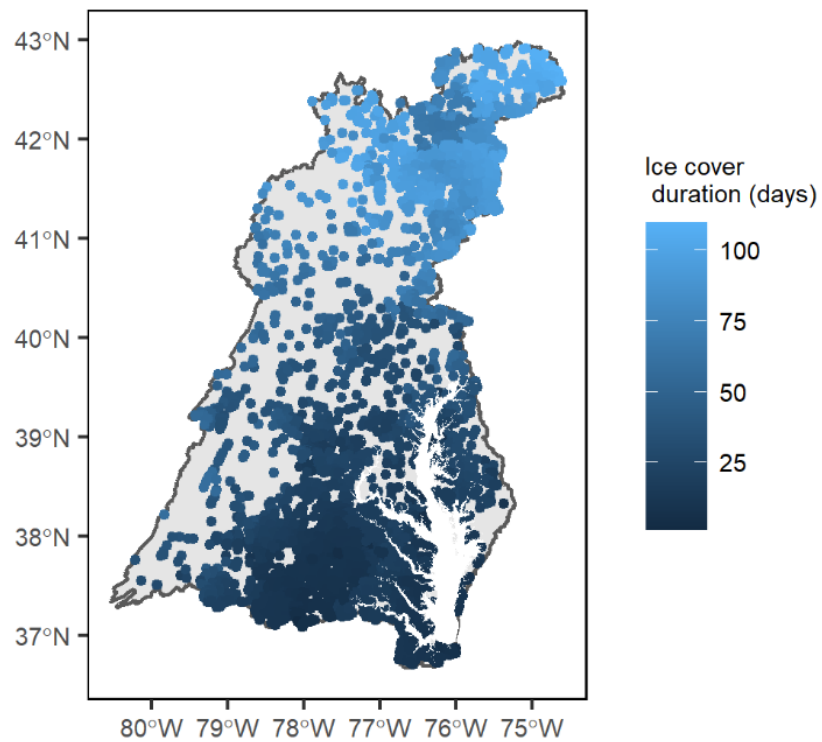
	Carbon Dioxide (CO ₂)	Methane (CH ₄)	Nitrous Oxide (N ₂ O)
Annual Emissions Reduction <i>Metric tons of each gas per year</i>	7,871 [6,646; 9,123]	2,042 [1,199; 3,158]	2.6 [1.2; 4.2]
Social Cost of Greenhouse Gas (SC-GHG) for 2025 Emissions* <i>2020\$ per metric ton of each gas 2.0% near-term discount rate</i>	\$212	\$2,025	\$54,139
Climate Benefits in 2025 <i>Millions of 2020\$</i>	\$1.7 [\$1.4; \$1.9]	\$4.1 [\$2.4; \$6.4]	\$0.14 [\$0.07; \$0.23]
Climate Benefits in 2075 <i>Millions of 2020\$</i>	\$3.1 [\$2.6; \$3.6]	\$13.0 [\$7.6; \$20.0]	\$0.32 [\$0.15; \$0.52]
Net Present Value of Climate Benefits in first 50 years <i>Millions of 2020\$</i>	\$73.8 [\$62.3; \$85.5]	\$253.0 [\$148.6; \$391.3]	\$6.9 [\$3.2; \$11.2]

Table 2 | The climate benefits of nutrient management. The monetized climate benefits of the Chesapeake Bay TMDL over the first 50 years of implementation (2025 to 2075). Values presented in brackets are based on the 95% confidence interval in the estimated annual emissions reductions only. *The SC-GHG changes over time and the 2025 value for each gas is presented as a reference point; the full time-series can be found in EPA (2023a)²², Table A.5.1.

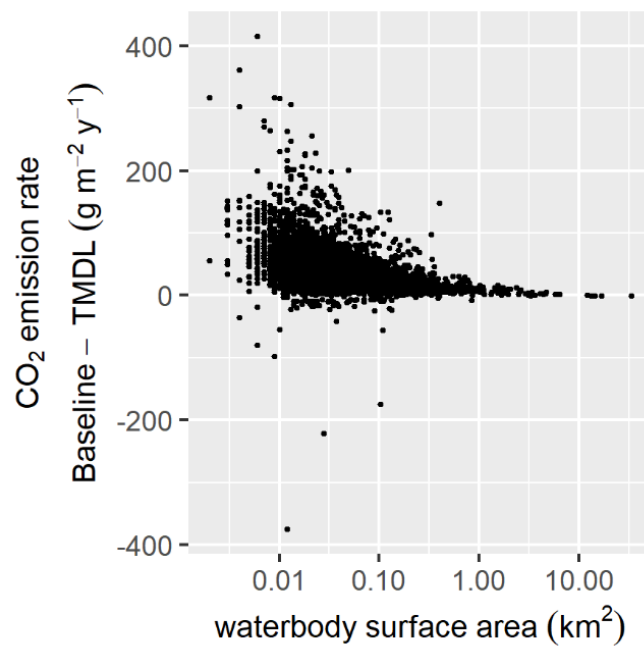
Extended Data Figures



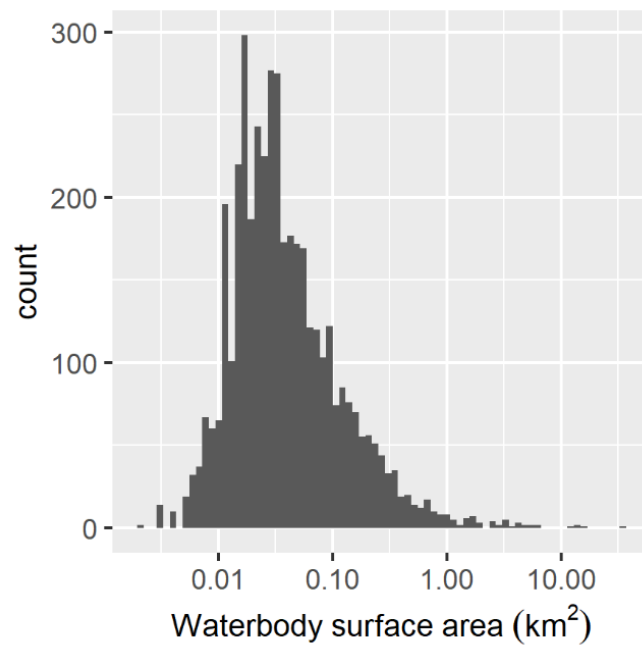
Extended Data Fig. 1 | The difference in pollutant concentrations and areal emissions rates from baseline to TMDL across the Chesapeake Bay Watershed. Color ramp reflects predicted values from Trend Surface models fit to each response variable. Black dots represent the location of lakes in the watershed. All variables were statistically significant and assessed at the $p=0.05$ level, and the final models were used to predict each response variable throughout a grid spanning the spatial domain of the watershed. The results suggest that there is a spatial pattern underlying the data that trends from the north-west of the region to the south-east. This is consistent with the spatial correlations of land use and ice cover that underpin our analysis.



Extended Data Fig. 2 | Number of days per year that waterbodies are ice covered. The days of ice-cover are used in the simulation to determine the amount of methane that accumulates in the waterbody during periods of ice cover.



Extended Data Fig. 3 | TMDL policy effect on areal CO₂ emission rates by waterbody size.



Extended Data Fig. 4 | Distribution of lake and reservoir surface areas in the Chesapeake Bay Watershed.

N = 4,221	Mean	Median	Inner 99th percentile	
Lake surface area (km ²)	0.116	0.031	0.005	1.33
Chl-a baseline (ug L ⁻¹)	33.5	20.8	0.05	115
Chl-a TMDL (ug L ⁻¹)	26.3	16.5	0.04	90.4
TP baseline (ug L ⁻¹)	86.4	44.1	0.06	267
TP TMDL (ug L ⁻¹)	68.2	34.6	0.05	213

Extended Data Table 1 | Surface area of lakes and reservoirs in the Chesapeake Bay Watershed. Chlorophyll-a (Chl-a) and total phosphorus (TP) are presented for TMDL and Baseline scenarios.

Extended Data Table 2 | Social Cost of Greenhouse Gas Emission Estimates 2025-2075.

Year	SC-CO ₂ (\$/mtCO ₂)	SC-CH ₄ (\$/mtCH ₄)	SC-N ₂ O (\$/mtN ₂ O)
2025	212	2,025	54,139
2026	215	2,101	55,364
2027	219	2,176	56,590
2028	223	2,252	57,816
2029	226	2,327	59,041
2030	230	2,403	60,267
2031	234	2,490	61,492
2032	237	2,578	62,718
2033	241	2,666	63,944
2034	245	2,754	65,169
2035	248	2,842	66,395
2036	252	2,929	67,645
2037	256	3,017	68,895
2038	259	3,105	70,145
2039	263	3,193	71,394
2040	267	3,280	72,644
2041	271	3,375	73,894
2042	275	3,471	75,144
2043	279	3,566	76,394
2044	283	3,661	77,644
2045	287	3,756	78,894
2046	291	3,851	80,304
2047	296	3,946	81,714
2048	300	4,041	83,124
2049	304	4,136	84,535
2050	308	4,231	85,945
2051	312	4,320	87,355
2052	315	4,409	88,765
2053	319	4,497	90,176
2054	323	4,586	91,586
2055	326	4,675	99,612
2056	330	4,763	100,935
2057	334	4,852	102,258
2058	338	4,941	103,581
2059	341	5,029	104,904
2060	345	5,118	106,227

2061	348	5,199	107,385
2062	351	5,280	108,542
2063	354	5,361	109,700
2064	357	5,442	110,857
2065	360	5,523	112,015
2066	363	5,604	113,172
2067	366	5,685	114,330
2068	369	5,765	115,487
2069	372	5,846	116,645
2070	375	5,927	117,802
2071	378	6,013	119,027
2072	382	6,099	120,252
2073	385	6,184	121,477
2074	388	6,270	122,702
2075	391	6,355	123,926

The social cost of greenhouse gas in 2020 USD per metric ton (\$/mt) based on a near-term discount rate of 2%(U.S. EPA 2023d).

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